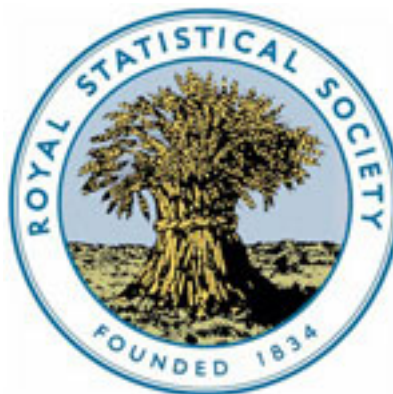


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# Scaling unobservable constructs in social science

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**Summary.** The use of the logit–probit model for constructing social measurement scales is explored by using two data sets. The first was specifically designed for scaling and consists of answers by 1490 individuals to five questions on economic self-determination. The second is from the Workplace Industrial Relations Survey and relates to answers to six questions addressed to 1005 firms about innovations. It is shown that although scales can be constructed their precision (reliability) is likely to be low. Further, for examples like these, latent class models may give an equally satisfactory account of the data.

**Keywords:** Attitude scaling; Factor analysis; Latent class models; Latent trait models; Latent variable models; Logit–probit model; Measurement

## 1. Social measurement

The scientific study of social phenomena depends, crucially, on the possibility of measuring the quantities which occur in social discourse. In some situations this can be done in a simple and direct manner. For example, sample survey or census data are capable of yielding estimates of things like the birth-rate, the proportion of households owning a piano, the average income of 18-year-old people or the proportion of school-children who have tried cannabis. There may be practical problems of coverage, measurement error or non-response and there may be room for argument about the relevance of any such measure for a particular purpose, but there is little doubt about the reality of what is being measured.

In contrast with this there are what may be described as *unobservable constructs*. These include abilities and attitudes, which individuals are supposed to have to a greater or lesser degree, such as quality of life, intelligence, aggressiveness, alienation, business confidence and conservatism. In social theorizing these are commonly treated just like other, directly measurable, variables in that their value is assumed to influence, and be influenced by, other variables. Indeed social science would be seriously impoverished if all such constructs were ruled out of court.

In response to this situation there has been a long tradition in the social sciences directed to making such *latent* variables operational. This goes back at least to the beginning of the 20th century when Charles Spearman invented factor analysis. Some of the history of that particular line of development is traced by Bartholomew (1995a) who noted that statisticians have often remained aloof from the enterprise with loss to all parties. Much of the work on constructing scales for unobservable constructs has coalesced into what is now known as psychometrics. This is largely focused on the measurement of human abilities, especially in educational testing. This is now a highly developed field with considerable commercial ramifications. The literature of the subject is enormous. Lord and Novick (1968) is an important early landmark and a more recent

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introduction to the field has been given by Hambleton *et al.* (1991). Several journals, such as *Psychometrics*, *Applied Social Measurement*, *Multivariate Behavioural Research and Educational and Psychological Measurement*, provide outlets for much of the continuing research in the field.

Inevitably, attempts to measure such ideologically sensitive things as intelligence and, especially, to make intergroup comparisons have generated heated debate. But, in spite of that, many psychometric scales are well established and widely used.

When we move from abilities to attitudes the situation is much less well developed. Relatively few established scales of measurement exist and there are major problems in the way of constructing adequate measures. The purpose of this paper is to explore the implications of this claim by empirical and theoretical means. We shall show, from a statistical perspective, why it is that valid and reliable scales are not easy to come by in the social sciences. This essentially pessimistic conclusion is offered in a constructive rather than destructive spirit. It is intended to encourage a shift of emphasis and resources in social research towards the measurement problem. Even if the obstacles cannot be completely overcome, it is possible, at a price, to put social measurement on a firmer footing.

We first describe briefly two data sets that are used to illustrate the approach. Then we outline the theoretical framework within which the measurement problem can be formulated based on Bartholomew (1987, 1996). Next we analyse the two data sets in detail. These analyses are typical of many in a much wider range which have been carried out and they illustrate the kind of obstacles to be overcome. The paper concludes with a discussion of the implications of the investigation for the measurement of unobservable constructs.

## 2. The two data sets

The first example is taken from an extensive study of social life feelings reported in Schuessler (1982) and Krebs and Schuessler (1987). The aim was to establish scales for use in social research of quality comparable with those used in ability testing. A substantial amount of work was needed to devise and test potential items. These took the form of propositions with which the respondent was invited to agree or disagree (sometimes more than two categories were provided, but these were subsequently reduced to binary items). Apart from the two publications mentioned above, analyses based on these data have been given in Bartholomew *et al.* (1996), Bartholomew and de Menezes (1996), Bartholomew and Schuessler (1991) and Bartholomew *et al.* (1993). For illustration we take here the example of scale 10: 'Economic self-determination'. We use the 'intercultural' version administered to samples of Germans and Americans and we present the data from the German sample consisting of 1490 individuals. This is one of the shorter scales with only five propositions. The more typical length is eight or nine items, with some scales going up to 12 items. We have chosen a short scale in the interests of economy, because tables of results require  $2^p$  rows, where  $p$  is the number of items. The propositions were as follows.

1. Anyone can raise his standard of living if he is willing to work at it.
2. Our country has too many poor people who can do little to raise their standard of living.
3. Individuals are poor because of the lack of effort on their part.
4. Poor people could improve their lot if they tried.
5. Most people have a good deal of freedom in deciding how to live.'

The most economical way to set out the data is as a list of the possible response patterns accompanied by their frequencies. To save repetition, we have combined this with some of the results of the analysis in Table 2 in Section 6. The responses are listed in the vector  $\mathbf{x}$  in the same order as the questions above.

The second example is from the Workplace Industrial Relations Survey (WIRS) of 1990. This was not designed for the construction of a scale, and the interest here is in seeing whether general purpose survey questions can be made to yield scales. The data arise from a series of questions about consultation in the workplace. The questions were addressed to firms and the series that we shall consider is as follows.

'Please consider the most recent change involving the introduction of new plant, machinery and equipment. Were discussions or consultations of any of the type on this card held either about the introduction of the change or about the way it was to be implemented?

1. informal discussion with individual workers
2. meeting with groups of workers
3. discussions in established joint consultative committee
4. discussions in specially constituted committee to consider the change
5. discussion with unions representatives at the establishment
6. discussions with paid union officials from outside.'

One might enquire whether the responses are indicative of a single underlying dimension or whether more than one is required. If these dimensions can then be identified with scales of substantive interest, we could proceed to use them in further explorations of the area.

Separate surveys were carried out for predominantly manual and non-manual groups and in relation to the introduction of new machinery or to changes in working practice. The largest subsample in this fourfold classification was for the non-manual workers on the introduction of new machinery, for which 1005 responses were obtained. We use these data, given in Table 1, for illustration but the other subsamples led to similar conclusions. Again, the ordering of the responses is the same as in the list of questions.

**Table 1.** Observed frequencies for each response pattern *x* for the WIRS data

| <i>Response pattern x</i> | <i>Frequency</i> | <i>Response pattern x</i> | <i>Frequency</i> | <i>Response pattern x</i> | <i>Frequency</i> |
|---------------------------|------------------|---------------------------|------------------|---------------------------|------------------|
| 000000                    | 132              | 000111                    | 0                | 111100                    | 3                |
| 000001                    | 5                | 001011                    | 2                | 111010                    | 6                |
| 000010                    | 17               | 010011                    | 11               | 110110                    | 4                |
| 000100                    | 13               | 100011                    | 10               | 101110                    | 8                |
| 001000                    | 12               | 001101                    | 0                | 011110                    | 35               |
| 010000                    | 172              | 010101                    | 1                | 111001                    | 1                |
| 100000                    | 65               | 100101                    | 1                | 110101                    | 0                |
| 000011                    | 0                | 011001                    | 2                | 101101                    | 1                |
| 000101                    | 0                | 101001                    | 8                | 011101                    | 3                |
| 001001                    | 1                | 110001                    | 10               | 110011                    | 3                |
| 010001                    | 4                | 111000                    | 13               | 101011                    | 4                |
| 100001                    | 11               | 110100                    | 5                | 011011                    | 11               |
| 000110                    | 0                | 101100                    | 2                | 100111                    | 2                |
| 001010                    | 2                | 011100                    | 13               | 010111                    | 8                |
| 010010                    | 38               | 110010                    | 26               | 001111                    | 0                |
| 100010                    | 34               | 101010                    | 31               | 111110                    | 4                |
| 001100                    | 1                | 011010                    | 28               | 111101                    | 2                |
| 010100                    | 45               | 100110                    | 5                | 111011                    | 5                |
| 100100                    | 20               | 010110                    | 22               | 110111                    | 2                |
| 011000                    | 21               | 001110                    | 1                | 101111                    | 7                |
| 101000                    | 24               |                           |                  | 011111                    | 30               |
| 110000                    | 55               |                           |                  | 111111                    | 3                |
|                           |                  |                           |                  | Total                     | 1005             |

### 3. A framework for measurement of unobserved constructs

Much of the confusion which permeates the scaling of unobserved constructs stems from a failure to see the problem in a sufficiently general framework. We begin by setting up the theoretical framework in a sufficiently abstract way to expose the essential structure of the problem. This has been done many times before and is included here, in outline, for the convenience of readers who are unfamiliar with the field.

Let us denote the latent construct by  $z$  which we assume to be continuous on some interval. As we cannot observe  $z$  directly we must infer something about its value from other variables on which it is believed to exert some influence. These are called *indicators* and we denote them by  $\mathbf{x} = (x_1, x_2, \dots, x_p)$ . If we suppose that there is some population of individuals on which  $\mathbf{x}$  may be observed, then  $\mathbf{x}$  will have some  $p$ -variate distribution;  $z$  will likewise vary in the population. The measurement problem to be addressed is, then, if we observe the value for  $\mathbf{x}$ , what can we say about the unobserved  $z$  for that individual? We denote the joint distribution of  $\mathbf{x}$  and  $z$  by  $f(\mathbf{x}, z)$ . All the information about  $z$  given  $\mathbf{x}$  is then contained in the *posterior* distribution with density  $f(z|\mathbf{x})$ . If this distribution could be estimated, we could summarize what we know about  $z$  by, for example  $E(z|\mathbf{x})$  and  $\text{var}(z|\mathbf{x})$ . The former would give us a *prediction* of  $z$  and  $\text{var}(z|\mathbf{x})$  (or its square root) would be a measure of its precision (which is often called *reliability* in this context).

According to Bayes's theorem

$$f(z|\mathbf{x}) \propto f(z)f(\mathbf{x}|z). \quad (1)$$

The theoretical approach to which we have referred is concerned with how to choose the distributions on the right-hand side; see Bartholomew (1987) for details. The analysis given there leads to three key results as follows.

- (a) The choice of the prior distribution  $f(z)$  is essentially arbitrary. Without loss of generality it may be chosen to have mean 0 and variance 1. As a matter of *convention* it is usually assumed to be normal.
- (b) If the interdependence of the  $x$ s is attributable solely to their common dependence on  $z$  then, if  $z$  is fixed, the  $x$ s will be mutually independent (the so-called postulate of *local* or *conditional* independence).
- (c) If the distribution of  $x_i$  given  $z$  belongs to the one-parameter exponential family (as in many common situations) then there is a single sufficient statistic for  $z$ . In other words, the posterior distribution of  $z$  given  $\mathbf{x}$  depends on  $\mathbf{x}$  only through a single function of  $\mathbf{x}$ . Furthermore, this statistic is often linear in the  $x$ s.

In other words all the information in  $\mathbf{x}$  about  $z$  in such cases is contained in a linear combination of the form

$$X = \sum_{i=1}^p \alpha_i x_i.$$

This is a remarkable result and is particularly useful in view of the strong intuitive feeling which many researchers have that sums, weighted or otherwise, are a suitable way of combining indicators to form an index. A further result shows that  $X$  is a monotonic function of  $E(z|\mathbf{x})$  so that any ranking of individuals based on  $X$  is the same as a ranking based on  $E(z|\mathbf{x})$ . Chang and Stout (1993) have shown that the posterior distribution  $f(z|\mathbf{x})$  tends to normality as  $p$  increases and this result, coupled with a result of Albanese and Knott (1993), establishes that, under the same limiting conditions,  $E(z|\mathbf{x})$  is a linear function of  $X$ . For all practical purposes we may, therefore, use  $X$  as a proxy for the unobservable  $z$ . Hence,

provided that we can obtain estimates of whatever unknown parameters the  $\alpha$ s may depend on, the measurement problem appears to be solved, at least in those cases where the conditional distributions of the  $x$ s, given  $z$ , belong to the exponential family.

It is when we implement this strategy that the practical difficulties appear. These are sufficiently illustrated by the case when all the  $x$ s are binary variables and therefore, for brevity, we confine the remainder of the discussion to that case.

#### 4. The logit–probit model

We suppose that  $x_i$  is a Bernoulli random variable with

$$\begin{aligned}\Pr\{x_i = 1|z\} &= \pi_i(z), \\ \Pr\{x_i = 0|z\} &= 1 - \pi_i(z)\end{aligned}\quad (2)$$

( $i = 1, 2, \dots, p$ ). This enables us to write

$$f(x_i|z) = \{1 - \pi_i(z)\} \exp[x_i \logit\{\pi_i(z)\}] \quad (3)$$

which is a member of the exponential family. The sufficiency property follows if we choose

$$\logit\{\pi_i(z)\} = \alpha_{i0} + \alpha_{i1}z \quad (i = 1, 2, \dots, p). \quad (4)$$

This, coupled with the assumed standard normal distribution, constitutes the *logit–probit* model. The sufficient statistic is easily shown to be

$$X = \sum_{i=1}^p \alpha_{i1}x_i.$$

A common variant is the probit–probit model in which the logit function is replaced by the normal distribution function. As

$$\logit(u) \doteq (\pi/\sqrt{3}) \Phi^{-1}(u)$$

it is clear that the two models are very close. The values of  $\alpha_{i0}$  and  $\alpha_{i1}$  for the probit–probit model will be less than those for the logit–probit model by a factor of about  $\sqrt{3}/\pi$ . The probit–probit model may also be derived by supposing that the value of each  $x$  is determined by whether or not an underlying continuous normal indicator exceeds some threshold. In that case, the underlying variable can be shown to satisfy a standard normal factor model. The logit–probit model is somewhat easier to fit and has the advantage that properties like the linearity of  $X$  and its sufficiency are exact. In practice the difference is negligible.

The logit–probit model may be fitted by maximum likelihood using the routines G11SAF and G11SBF in the Numerical Algorithms Group (NAG) library (Numerical Algorithms Group, 1987) or by TWOMISS (Albanese and Knott, 1990). Approximate estimates may be obtained by using the probit–probit model fitted using LISREL (Jöreskog and Sörbom, 1993), LISCOMP (Muthén, 1987) or TESTFACT (Wilson *et al.*, 1991) though the last three give no information about the sampling behaviour of the estimates.

#### 5. Goodness of fit

The obvious way to judge the goodness of fit of the model is to compare the observed and expected frequency distributions across the response patterns. Provided that  $n$ , the sample

size, is large compared with  $2^p$  (the number of patterns) this can be done with the usual  $\chi^2$  or  $G^2$  (log-likelihood ratio test). Unless  $n \gg 2^p$  there will usually be some very small expected frequencies and this will necessitate the pooling of categories. The exact sampling distribution of  $\chi^2$  and  $G^2$  has been investigated by Bartholomew and Tzamourani (1997) who showed that the  $\chi^2$ -approximation to the distribution tends to overstate the true significance, but the examples presented here are within the range where the usual test can be trusted.

A global test of fit is a rather blunt instrument and it is usually more informative to examine the nature of any departure from the null hypothesis. This may be done by looking at the individual contributions to  $\chi^2$ , but a more informative way is to proceed as follows.

The  $2^p$  response probabilities  $f(\mathbf{x})$  are equivalent to the following set of marginal probabilities.

$$\begin{aligned} \Pr\{x_i = 1\} & \quad (i = 1, 2, \dots, p), \\ \Pr\{x_i = 1, x_j = 1\} & \quad (i, j = 1, 2, \dots, p; i \neq j), \\ \Pr\{x_i = 1, x_j = 1, x_k = 1\} & \quad (i, j, k = 1, 2, \dots, p; i \neq j \neq k), \\ & \vdots \\ \Pr\{x_1 = 1, x_2 = 1, \dots, x_p = 1\}. \end{aligned} \quad (5)$$

This can be seen by noting that the number of probabilities listed in expression (5) is

$$\binom{p}{1} + \binom{p}{2} + \binom{p}{3} + \dots + \binom{p}{p} = 2^p - 1$$

which is equal to the number of independent response probabilities. The particular interest of this set of marginal probabilities is that they describe different aspects of the interdependence of the  $x$ s.

The probabilities of the form  $\Pr\{x_i = 1\}$  are the first-order margins of the joint distribution and they tell us nothing about the associations among the  $x$ s. The bivariate probabilities, or second-order margins, describe the pairwise associations. We can, of course, construct the full bivariate distribution of  $x_i$  and  $x_j$  by using the first-order margins. For example

$$\Pr\{x_i = 0, x_j = 1\} = \Pr\{x_j = 1\} - \Pr\{x_i = 1, x_j = 1\}. \quad (6)$$

All the third-order margins can be determined from  $\Pr\{x_i = 1, x_j = 1, x_k = 1\}$  by using the second- and first-order margins in a similar fashion, and so on. The program TWOMISS calculates all margins up to order 3 and we make use of these in the analysis which follows.

For any marginal frequency we define a residual  $R = (O - E)^2/E$  where  $O$  and  $E$  are the observed and expected frequencies for the margin in question. Earlier examples of the use of such residuals in judging goodness of fit are found in Knott and Tzamourani (1996), Bartholomew *et al.* (1996) and Bartholomew and de Menezes (1996). The set of such residuals for margins of a given order will not be independent but individually they will be approximately distributed like  $\chi^2$  with 1 degree of freedom. Typically, if the fit is good, we should expect them to be about 1; values larger than 4, say, would alert us to possible deviations from the model.

## 6. Fitting the model

We have proposed a model, with attractive statistical properties, which provides a basis for arriving at scales for unobservable constructs. It shows that if we can find a set of  $p$  binary items,  $\mathbf{x}$ , such that model (4) fits data sampled at random from some population of interest then we may predict an individual's position on the scale of  $z$  by using some measure of location of the posterior distribution such as  $E(z|\mathbf{x})$ . We can determine the precision of that prediction by using the posterior standard deviation  $\sigma(z|\mathbf{x})$ . If we compare the latter with the prior standard deviation of 1 we shall see by how much the knowledge of  $\mathbf{x}$  reduces our uncertainty about  $z$ . In constructing a scale, therefore, our aim must be to find a set of items whose joint distribution is adequately described by the logit–probit model and which provides reasonably precise predictions of the latent scale position. In this section we illustrate the implementation of this strategy on the two examples described in Section 2.

### 6.1. Social life feelings data

The data, together with a range of derived statistics, are given in Table 2. The 32 distinct response patterns are in the last column. They are listed in order of the scale values in the third and fifth columns. The frequencies are in the first column. It is immediately clear that some response patterns occur much more frequently than others. The purpose of fitting the model is to see whether this pattern can be explained by the logit–probit model. The estimated parameter values are given in Table 3, together with their asymptotic standard errors. The  $\chi^2$  goodness-of-fit statistic is 32.87 and the log-likelihood ratio statistic ( $G^2$ ) is 35.01, both on 16 degrees of freedom. The number of degrees of freedom would ordinarily be  $32 - 1$  (total)  $- 10$  (number of parameters)  $= 21$  but some categories, with small expected frequencies, have been pooled.

The standard errors, especially of the  $\alpha_{1i}$ , are quite large and appear to be roughly proportional to the estimated values. Even so, all the  $\alpha$ s are clearly positive. The goodness of fit, judged by either  $\chi^2$  or  $G^2$ , is questionable ( $P$  is close to 0.01). An inspection of the observed and expected frequencies columns shows that agreement in general is very good, and that most of the deviation from the model is attributable to a few response patterns. Monte Carlo studies (see, for example, Bartholomew and Tzamourani (1997)) have shown that  $\chi^2$  and  $G^2$  tend to overestimate the level of significance. More relevant, however, is the comparison of the observed and expected frequencies for the two- and three-way margins. For almost all pairwise residuals,  $(O - E)^2/E$  is less than 1. The largest values are 2.68 and 2.10 which are still quite small. The three-way margins likewise yield no residual greater than 2.20, so we conclude that all two- and three-way associations are well predicted by the model.

This evidence seems a sufficient basis for constructing a scale from these items, and we return to Table 2 for the scaling and its properties. The predicted scale values in the third column are on a standard normal scale. As we expected, these show a close correlation with the sufficient statistic  $X$ ; in fact a plot would show the relationship to be almost linear.

It is interesting to compare the weighted  $X$ -score with the unweighted sum  $\sum x_i$  given in the adjacent column. This is the score that we would use if we decided simply to count the positive answers. Broadly, the correspondence is close, but it is instructive to look at cases which are out of line. The response pattern 00010, for example, scores substantially higher than the other cases where there was only one positive response. This is because item 4 has a much higher weight than the others (2.55), and so a single positive response on that item is worth more than two positive responses on, for example, items 1 and 2. The advantage of the model, over the 'commonsense' scale obtained by adding up positive answers, is that it allows

**Table 2.** Observed (*O*) and expected (*E*) frequencies, posterior means and standard deviations and values of the sufficient statistic *X* for each response pattern **x** for the logit–probit model fitted to the social life feelings data

| <i>O</i> | <i>E</i> | $E(y X)$ | $\sigma(y X)$ | <i>X</i> | $\sum x_i$ | <b>x</b> |
|----------|----------|----------|---------------|----------|------------|----------|
| 156      | 162.0    | −1.19    | 0.72          | 0.00     | 0          | 00000    |
| 174      | 174.0    | −0.85    | 0.67          | 0.71     | 1          | 01000    |
| 26       | 22.2     | −0.76    | 0.66          | 0.92     | 1          | 00001    |
| 8        | 5.2      | −0.64    | 0.64          | 1.20     | 1          | 10000    |
| 127      | 122.3    | −0.50    | 0.63          | 1.53     | 1          | 00100    |
| 35       | 31.9     | −0.46    | 0.62          | 1.64     | 2          | 01001    |
| 8        | 8.1      | −0.36    | 0.61          | 1.91     | 2          | 11000    |
| 2        | 1.1      | −0.28    | 0.61          | 2.12     | 2          | 10001    |
| 208      | 208.7    | −0.23    | 0.60          | 2.24     | 2          | 01100    |
| 26       | 30.2     | −0.16    | 0.60          | 2.45     | 2          | 00101    |
| 14       | 16.9     | −0.12    | 0.59          | 2.55     | 1          | 00010    |
| 4        | 8.3      | −0.06    | 0.59          | 2.73     | 2          | 10100    |
| 2        | 2.3      | −0.02    | 0.59          | 2.83     | 3          | 11001    |
| 65       | 65.3     | 0.09     | 0.59          | 3.17     | 3          | 01101    |
| 36       | 37.6     | 0.12     | 0.58          | 3.26     | 2          | 01010    |
| 19       | 19.3     | 0.18     | 0.58          | 3.44     | 3          | 11100    |
| 9        | 5.8      | 0.19     | 0.58          | 3.47     | 2          | 00011    |
| 4        | 3.0      | 0.25     | 0.58          | 3.65     | 3          | 10101    |
| 1        | 1.8      | 0.29     | 0.58          | 3.74     | 2          | 10010    |
| 66       | 56.5     | 0.40     | 0.59          | 4.08     | 2          | 00110    |
| 13       | 16.2     | 0.44     | 0.59          | 4.19     | 3          | 01011    |
| 10       | 8.8      | 0.50     | 0.59          | 4.36     | 4          | 11101    |
| 5        | 5.3      | 0.53     | 0.59          | 4.46     | 3          | 11010    |
| 3        | 0.9      | 0.61     | 0.60          | 4.67     | 3          | 10011    |
| 195      | 182.7    | 0.65     | 0.60          | 4.79     | 3          | 01110    |
| 16       | 31.8     | 0.73     | 0.61          | 5.00     | 3          | 00111    |
| 18       | 11.2     | 0.83     | 0.62          | 5.27     | 3          | 10110    |
| 3        | 3.4      | 0.88     | 0.62          | 5.38     | 4          | 11011    |
| 129      | 130.9    | 1.01     | 0.64          | 5.72     | 4          | 01111    |
| 31       | 49.9     | 1.12     | 0.65          | 5.99     | 4          | 11110    |
| 9        | 9.6      | 1.21     | 0.66          | 6.20     | 4          | 10111    |
| 68       | 56.7     | 1.54     | 0.70          | 6.91     | 5          | 11111    |

**Table 3.** Parameter estimates and asymptotic standard errors (in parentheses) for the logit–probit model fitted to the social life feelings data

| <i>Item</i> | $\alpha_{i0}$ | $\alpha_{i1}$ |
|-------------|---------------|---------------|
| 1           | −2.35 (0.13)  | 1.20 (0.14)   |
| 2           | 0.80 (0.06)   | 0.71 (0.09)   |
| 3           | 0.99 (0.09)   | 1.53 (0.17)   |
| 4           | −0.67 (0.13)  | 2.55 (0.41)   |
| 5           | −1.10 (0.07)  | 0.92 (0.10)   |

the data to give us a relative valuation of the importance of the questions in relation to the underlying attitude dimension.

The remaining item in Table 2 is the posterior standard deviation  $\sigma(y|X)$ . This is remarkably constant over all response patterns. The average value of the posterior variance was shown by Bartholomew (1995b) to be close to the standard test–retest coefficient of reliability used in test theory, but it is more informative to look at the individual values. The largest uncertainties are associated with the extreme response patterns, but all are large. The typical

standard deviation of around 0.60 must be compared with the unit standard deviation of the prior distribution; this shows that a precise measurement is impossible with these items.

Thus even though we have a satisfactorily fitting model, which produces scores which behave in the expected way, the scale values are subject to considerable uncertainty. Extensive analyses of many similar sets of data from the social life feelings scales confirm this conclusion. The overall precision depends on both the number of questions and the magnitude of the  $\alpha_{ij}$ s. Increasing the number of questions increases the precision, but there are serious practical limitations on how many questions can be included. Whereas in educational tests 50 or more test items may be used, it is often difficult to devise more than a dozen questions that are indicative of a single attitude dimension. This places a severe practical limit on what can be achieved. The magnitude of an individual  $\alpha_{i1}$  measures the importance of the item in determining the latent scale. (They are essentially the same as factor loadings as that term is used in factor analysis and we occasionally describe them thus.) A few items with large  $\alpha$ s will thus give a better scale than more items with small  $\alpha$ s. Those estimated in this example are fairly typical, but in a study of attitudes to abortion (see Knott *et al.* (1990)) much higher values were observed.

There is a second limitation which applies to all scales obtained in this way. Latent trait models (of which the logit–probit model is an example) are difficult to distinguish empirically from latent class models. Instead of supposing individuals to be distributed along a continuum, a latent class model supposes them to be in two or more distinct classes. Examples are given in Bartholomew and de Menezes (1996). When the data provide as much support for an allocation to classes as to a latent scale, our confidence in any such scale is reduced. The reason for this near lack of identifiability has been explained in Bartholomew (1987), p. 102, and Bartholomew (1996), p. 204. It can be regarded as another manifestation of the lack of precision found with the latent trait model.

It is tempting to argue that, if latent variable modelling provides such poor measuring instruments, the approach should be abandoned in favour of more traditional methods. However, this would be a misreading of the situation. The modelling of the measurement problem has merely made explicit what is already implicit in less formal methods. It has, for example, provided a theoretical framework justifying the use of linear combinations of indicators and has provided the means of judging their performance. Those who use such measures in a more intuitive way are, in fact, behaving as if they had adopted the latent variable framework, and they cannot escape the implication of their actions merely by ignoring this fact.

## 6.2. Management–worker consultation in firms

Unlike the items used in the social life feelings example, the question asked in the management–worker consultation was not designed for scale construction, and yet it deals with an issue which we might expect to be susceptible to scaling. In this respect it is similar to many questions that are used in sample surveys, and the point in considering it here is to see how successful the modelling approach of this paper is likely to be in such cases.

Most obviously, the question is concerned with the amount of consultation which takes place. Those firms who place a high value on consultation might be expected to use all or most methods. Others might use few or none. However, as we go down the list of alternatives, we move from informal to formal modes, and so we should be alert to the possibility that firms may also vary among themselves, according to the relative weight which they put on these two styles of consultation.

**Table 4.** Observed and predicted association between items 1 and 2 for the WIRS data†

| Item 2 | Item 1                             |                                    |
|--------|------------------------------------|------------------------------------|
|        | 1                                  | 0                                  |
| 1      | $O = 142, E = 221.8$<br>$R = 28.7$ | $O = 223, E = 153.2$<br>$R = 41.5$ |
| 0      | $O = 444, E = 364$<br>$R = 17.5$   | $O = 186, E = 265.7$<br>$R = 23.9$ |

† $O$ , observed frequency;  $E$ , expected frequency;  $R$ , residual.

The one-dimensional model is decisively rejected. We find  $G^2 = 269.4$  and  $\chi^2 = 264.2$  each on 32 degrees of freedom, after pooling where necessary, which indicates a highly significant departure from the model. The nature of the deviation is clear if we look at the observed and expected values for the two- and three-way margins. By far the largest discrepancies involve items 1 and 2 in the list of types of consultation. The model conspicuously fails to predict the pairwise association between these two items as Table 4 shows. Many fewer firms use both informal discussion with individuals and meetings with groups than the model predicts. Put another way, more adopt a ‘one but not both’ approach than we would expect.

We next fit the logit–probit model with two latent variables to see whether the introduction of a second factor can explain this discrepancy. The fit is still poor, though better than the one-factor model with  $\chi^2 = 133.6$  on 24 degrees of freedom. It is worth pausing before rejecting the model outright to look at the parameter estimates as given in Table 5. The positive and roughly equal loadings on the first factor support our intuition that firms will vary along a dimension corresponding to the *amount* of consultation which takes place. The bipolar character of the second factor suggests a second dimension in which the formal–informal mix varies. Given that this seemingly plausible interpretation is not supported by the overall goodness-of-fit test, we look at the residuals. None of the values of  $(O - E)^2/E$  for the two-way margins exceeds 0.44 so the model is extremely successful in predicting the pairwise associations between the items. However, if we turn to the three-way margins, a striking pattern emerges. This shows that the poor fit arises from those triples which include item 1. In every case the observed frequency is greater than expected. This suggests that item 1 is inappropriate in some way. It involves informal contact and is therefore inherently imprecise and potentially subject to varying interpretation by respondents. We therefore omitted this question and refitted the two-factor model.

The fit was much improved with  $G^2 = 35.1$  on 13 degrees of freedom. Although this is still significant, as judged by normal statistical criteria, the fit to the two-way margins is now

**Table 5.** Parameter estimates for the two-factor logit–probit model with standard errors (in parentheses) for the WIRS data

| Item | $\alpha_{i0}$ | $\alpha_{i1}$ | $\alpha_{i2}$ |
|------|---------------|---------------|---------------|
| 1    | 0.92 (0.11)   | 0.95 (0.16)   | −2.08 (0.27)  |
| 2    | 0.54 (0.09)   | 1.54 (0.19)   | −0.98 (0.14)  |
| 3    | −1.40 (0.09)  | 1.31 (0.14)   | 1.17 (0.13)   |
| 4    | −1.47 (0.09)  | 1.22 (0.12)   | 0.12 (0.12)   |
| 5    | −1.97 (0.10)  | 1.58 (0.17)   | 1.24 (0.14)   |
| 6    | −2.39 (0.11)  | 1.05 (0.12)   | 1.06 (0.13)   |

**Table 6.** Parameter estimates for the two-factor logit–probit model with standard errors (in parentheses) for the WIRS data with item 1 omitted

| <i>Item</i> | $\alpha_{i0}$ | $\alpha_{i1}$ | $\alpha_{i2}$ |
|-------------|---------------|---------------|---------------|
| 2           | 1.10 (0.20)   | 4.15 (0.47)   | −2.64 (0.34)  |
| 3           | −1.43 (0.10)  | 1.20 (0.14)   | 1.35 (0.16)   |
| 4           | −1.43 (0.09)  | 1.07 (0.11)   | 0.36 (0.13)   |
| 5           | −0.94 (0.10)  | 1.42 (0.15)   | 1.32 (0.16)   |
| 6           | −2.31 (0.11)  | 0.92 (0.12)   | 1.02 (0.14)   |

excellent and the three-way margins have no residual bigger than 1.61. We therefore have a model which accurately predicts the second- and third-order margins. The new parameter estimates are given in Table 6. The general factor persists but with item 2 more heavily weighted. The bipolar character of the second factor survives, with item 2 as the sole representative of the informal type of consultation. The whole analysis thus points to two sources of variation among firms. Firstly, there is the extent to which they take part in consultation at all and, secondly, the relative emphasis which they place on formal and informal channels.

If this interpretation is correct we should be able to use items 3–6 to construct a scale measuring the amount of formal consultation which takes place. Fitting the one-factor model to these items yields  $G^2 = 16.6$  ( $\chi^2 = 14.5$ ) on 7 degrees of freedom. All residuals for two-way margins are very small (less than 0.12) and no three-way residual exceeds 1. The weights, and their standard errors, are  $\alpha_{11} = 1.77$  (0.26),  $\alpha_{21} = 0.98$  (0.14),  $\alpha_{31} = 1.97$  (0.31) and  $\alpha_{41} = 1.37$  (0.19). Having regard to the size of the standard errors an adequate scaling, for most purposes, would be obtained by simply adding up the number of positive responses to the four items. This total score is consistent with the rank ordering based on  $E(z|\mathbf{x})$ .

As a final word of caution, note that the posterior standard deviations range from 0.59 for mixed response patterns to 0.75 for the extreme response 0000. These standard deviations are large indicating, as we would expect, that the latent location cannot be determined at all precisely on only four binary items.

In view of our remarks on the social life feelings data about the similarity of latent class and latent variable models, it would not be surprising to find the same situation here. This proves to be the case, though we need a model with five classes to achieve an adequate fit. This reflects the subtleties which our analysis of residuals has uncovered. Table 7 gives the parameter estimates for a five-class model. The last row gives the estimated proportionate

**Table 7.** Parameter estimates for a latent class model with five classes fitted to the WIRS data

| <i>Item</i>            | <i>Estimates for the following classes:</i> |           |            |           |          |
|------------------------|---------------------------------------------|-----------|------------|-----------|----------|
|                        | <i>I</i>                                    | <i>II</i> | <i>III</i> | <i>IV</i> | <i>V</i> |
| 1                      | 0.00                                        | 1.00      | 1.00       | 0.08      | 0.89     |
| 2                      | 0.43                                        | 0.47      | 1.00       | 0.01      | 1.00     |
| 3                      | 0.19                                        | 0.07      | 0.18       | 0.79      | 0.72     |
| 4                      | 0.14                                        | 0.08      | 0.40       | 0.28      | 0.61     |
| 5                      | 0.30                                        | 0.10      | 0.23       | 0.77      | 0.90     |
| 6                      | 0.14                                        | 0.03      | 0.00       | 0.31      | 0.45     |
| Proportion<br>in class | 0.29                                        | 0.34      | 0.13       | 0.07      | 0.17     |

sizes of the classes. Each column gives the probabilities that a firm in the latent class corresponding to that column will respond positively to the item represented by the row. At the two extremes are classes comprising firms who have high probabilities of using all or none of the methods of consultation. Of the remainder, class IV consists of firms who mainly use the more formal methods (items 3–6). Classes II and III use mainly the two informal methods and they are distinguished from one another principally by their response to item 2. In class III it is always used, but in class II it is used only by about half the firms. It should be noted that the latent class model has more parameters than the latent trait model and hence the latter might be preferred on grounds of parsimony.

We have included these summary results of the latent class analysis to emphasize that a satisfactory fit of a latent variable model does not establish the existence of ‘real’ underlying scales. It merely shows that the scales we construct by this means are capable of explaining the associations we find in the data. In the present example, the two models are consistent in that both identify item 1 as being somewhat anomalous and both draw attention to the distinction between formal and informal modes of consultation. Both analyses provide plausible accounts of firms’ behaviour in the matter of consultation.

## 7. Conclusions

Our aim has been to explore the possibility of constructing valid and reliable measurement scales for unobservable social variables. The latent variable approach to modelling, leading to the logit–probit model, provides a framework within which the measurement problem can be posed. Technically, the problem is solved, but practically there are serious limitations on what can be achieved. Although our demonstration of this conclusion rests largely on two examples, relating to binary data only, they are representative of a much wider body of experience, some of which has been referred to. In any case, the discovery of problems in only one example is sufficient to show that we must tread with care.

The economic self-determination scale was chosen because it was purpose built and had been the subject of much refinement before attempts to evaluate its performance by the present method were attempted. We have shown that there are good statistical reasons for treating these items as indicative of a single underlying attitudinal dimension. The posterior analysis reveals, however, the lack of precision (reliability) which is associated with the scale.

The data from the WIRS are more typical of what can be gleaned from social surveys. Here we had to address the prior question of whether the items reflected one latent dimension and, if not, how we might identify dimensions of interest and construct a scale that is appropriate for them. We showed that, in this example, it was possible to construct a scale, albeit unreliable, but its potential usefulness is a substantive rather than a statistical matter.

The most serious objection to our approach is that other models, in particular the latent class model, may give an equally good account of the data. It does not follow from this that methods of scaling which are not model based are to be preferred. On the contrary, we have shown that, in the context of binary data, any linear combination of the indicators can be regarded as a sufficient statistic for *some* logit–probit model. The theoretical framework that we have adopted makes explicit what would otherwise have been implicit. All of this makes it important to emphasize, again, that we are dealing with *constructs* rather than with ‘real’ variables. Such constructs are useful in so far as they correspond to our intuitive idea of the concept that we are trying to express numerically. Their merits can then be judged, pragmatically, by the extent to which they allow us to make predictions and by the precision and economy which they introduce into social theorizing.

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